

Ex NO 3 b.

DATA PREPROCESSING

Aim

To perform data preprocessing on a dataset using Python and Scikit-learn by handling missing values, encoding categorical data, splitting the dataset, and applying feature scaling.

Procedure

1. Import the required Python libraries.
2. Mount Google Drive and load the dataset using Pandas.
3. Display the first few records of the dataset.
4. Inspect the dataset using `df.info()` and `df.shape`.
5. Separate the independent variables (X) and dependent variable (Y).
6. Convert the independent variables into an array.
7. Identify and handle missing values using `SimpleImputer` with the mean strategy.
8. Encode the categorical Country column using `LabelEncoder`.
9. Apply One-Hot Encoding to convert categorical country values into dummy variables.
10. Encode the dependent variable Purchased using `LabelEncoder`.
11. Split the dataset into training and testing sets using `train_test_split`.
12. Apply `StandardScaler` for feature scaling.
13. Display the preprocessed training and testing datasets.

Program

```
# Step 1: Import libraries and load dataset
```

```
from google.colab import drive
```

```
drive.mount('/content/drive')
```

```
import pandas as pd
```

```
import numpy as np
```

```
df = pd.read_csv('/content/drive/MyDrive/Datasets/Data.csv')
```

```
# Display dataset
```

```
df.head()
```

```
# Step 2: Check dataset information
```

```
df.info()
```

```
print(df.shape)
```

```
# Step 3: Separate independent and dependent variables
```

```

x = df[['Country', 'Age', 'Salary']]

y = df[['Purchased']].values

# Convert X into array

x = df[['Country', 'Age', 'Salary']].values

# Step 4: Handle missing values

from sklearn.impute import SimpleImputer

imputer = SimpleImputer(

    missing_values=np.nan,

    strategy='mean')

imputer.fit(x[:, 1:3])

x[:, 1:3] = imputer.transform(x[:, 1:3])

print(x)

# Step 5: Encode categorical data

from sklearn.preprocessing import LabelEncoder

label_encoder_x = LabelEncoder()

x[:, 0] = label_encoder_x.fit_transform(x[:, 0])

print(x)

# Step 6: One-Hot Encoding

from sklearn.preprocessing import OneHotEncoder

onehotencoder = OneHotEncoder()

x_country = onehotencoder.fit_transform( df.Country.values.reshape(-1, 1)

).toarray()

print(x_country)

# Encode dependent variable

labelencoder_y = LabelEncoder()

```

```
y = labelencoder_y.fit_transform(y)

print(y)

# Step 7: Split dataset into training and testing sets

from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split( x, y, test_size=0.2, random_state=0)

print(x_train)

print(x_test)

print(y_train)

# Step 8: Feature Scaling

from sklearn.preprocessing import StandardScaler

sc_x = StandardScaler()

x_train = sc_x.fit_transform(x_train)

x_test = sc_x.transform(x_test)

print(x_train)

print(x_test)
```

Conclusion

Thus, the given dataset was successfully preprocessed by handling missing values, encoding categorical variables, splitting the data into training and testing sets, and performing feature scaling.